

CSA0501

DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENT

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EXPERIMENT – 14

AIM

To learn and implement **TCL commands COMMIT, SAVEPOINT and ROLLBACK** in MySQL.

Procedure

1. Open MySQL Workbench and connect to the MySQL server.
2. Select the BikeShowroomDB database.
3. Create a separate table for the transaction experiment.
4. Insert sample bike records.
5. Use COMMIT to permanently save the records.
6. Perform an UPDATE operation.
7. Create a SAVEPOINT.
8. Perform another INSERT operation.
9. Create another SAVEPOINT.
10. Use ROLLBACK TO SAVEPOINT to undo the changes after the selected savepoint.
11. Use SELECT to verify the final records.

QUESTIONS & OUTPUTS :

1. Create the table and insert bike records
2. Update a bike and create a savepoint
3. Insert another bike and create another savepoint

1. Create the table and insert bike records

```
1 • USE BikeShowroomDB;
2
3 • CREATE TABLE bike_transaction (
4     Bike_ID INT PRIMARY KEY,
5     Bike_Name VARCHAR(50),
6     Price DECIMAL(10,2)
7 );
8
9 • INSERT INTO bike_transaction
10 VALUES
11 (1, 'Apache RTR 200', 145000),
12 (2, 'Pulsar NS200', 165000),
13 (3, 'Classic 350', 225000);
14
```

Result Grid

Bike_ID	Bike_Name	Price
1	Apache RTR 200	145000.00
2	Pulsar NS200	165000.00
3	Classic 350	225000.00
NULL	NULL	NULL

bike_transaction 14 x

Output

Action Output

#	Time	Action	Message
1	18:27:04	USE BikeShowroomDB	0 row(s) affected
2	18:27:04	CREATE TABLE bike_transaction (Bike_ID INT PRIMARY KEY, Bike_Name VARCHAR(50), Price DECIMAL(10,2))	0 row(s) affected
3	18:27:04	INSERT INTO bike_transaction VALUES (1, 'Apache RTR 200', 145000), (2, 'Pulsar NS200', 165000), (3, 'Classic 350', 225000)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
4	18:27:04	COMMIT	0 row(s) affected
5	18:27:04	SELECT * FROM bike_transaction LIMIT 0, 1000	3 row(s) returned

2. Update a bike and create a savepoint

```
1 • UPDATE bike_transaction
2 SET Price = 150000
3 WHERE Bike_ID = 1;
4
5 • SAVEPOINT A;
6 • SELECT * FROM bike_transaction;
```

Result Grid

Bike_ID	Bike_Name	Price
1	Apache RTR 200	150000.00
2	Pulsar NS200	165000.00
3	Classic 350	225000.00
NULL	NULL	NULL

bike_transaction 15 x

Output

Action Output

#	Time	Action	Message
1	18:27:44	UPDATE bike_transaction SET Price = 150000 WHERE Bike_ID = 1	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
2	18:27:44	SAVEPOINT A	0 row(s) affected
3	18:27:44	SELECT * FROM bike_transaction LIMIT 0, 1000	3 row(s) returned

3. Insert another bike and create another savepoint

```
1 • INSERT INTO bike_transaction
2 VA Execute the selected portion of the script or everything, if there is no selection
3
4 • SAVEPOINT B;
5 • SELECT * FROM bike_transaction;
```

Result Grid			Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
Bike_ID	Bike_Name	Price				
1	Apache RTR 200	150000.00				
2	Pulsar NS200	165000.00				
3	Classic 350	225000.00				
4	R15 V4	190000.00				
* NULL	NULL	NULL				

bike_transaction 16 x Apply Rev

Output

#	Time	Action	Message
✓ 1	18:28:26	INSERT INTO bike_transaction VALUES (4, 'R15 V4', 190000)	1 row(s) affected
✓ 2	18:28:26	SAVEPOINT B	0 row(s) affected
✓ 3	18:28:26	SELECT * FROM bike_transaction LIMIT 0, 1000	4 row(s) returned

RESULT

Thus, the **COMMIT**, **SAVEPOINT** and **ROLLBACK** TCL commands were successfully implemented